Aditya S. Mate

Curriculum Vitae

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Summary

Applied Scientist at **Microsoft** and **Harvard PhD** (computer science) with **8 years of AI/ML research** experience. Research focuses on LLMs, reasoning agents, evaluation, safety, routing and decision-making under uncertainty. Prior research experience at **Google Research** and **IBM Research**. Published at NeurIPS, ICML, AAAI, AAMAS, EMNLP etc. with experience translating research into deployed ML systems in product settings.

Education

2018–2023 **Harvard University**, Cambridge, MA
Ph.D. in Computer Science | Thesis: Actualizing Impact of AI and Optimization in Public Health.

2013–2018 **Indian Institute of Technology Bombay**, Mumbai
B.Tech & M.Tech in Electrical Engineering | Minor: Computer Science and Engineering.

Industry Experience

July'23–Present **Microsoft**, Cambridge, MA.
APPLIED SCIENTIST 2 | Manager: Dr. Yingnong Dang

- Currently focus on developing probabilistic models and LLM reasoning agents within Azure's AI Model & Solution team to improve the quality and reliability of Azure services.
- Selected at Microsoft's AI Incubation Program by the Office of CTO, to lead infusion of AI/ML research in product problems. Collaborate with four product organizations across Microsoft: Strategic Missions and Technologies, Microsoft 365, GenAI, and Microsoft Search, Assistant and Intelligence (MSAI).
- Led & mentored research and developed ML/LLM solutions across LLM evaluation, fine-tuning and safety, cost-aware LLM routing, RAG/Graph RAG, and reasoning over time-series data, translating research into deployed ML systems in product settings.

2021–2022 **Google Research**, United States / India.
RESEARCH INTERN / STUDENT RESEARCHER | Managers: Philip Nelson, Dr. Aparna Taneja

- Led the development of a novel approach for evaluating algorithmic resource-allocation policies in randomized trials, enabling more accurate causal estimates; published at **ICML 2023**.
- Led the field study and deployment of a restless multi-armed bandit system for improving maternal and child health outcomes, translating ML research into a real-world intervention program; published at **AAAI 2022** and recognized with an **IAAI 2023 Innovative Application Award**.
  Awarded a peer bonus for "tremendous efforts and contributions" during the internship.
- Contributed to and mentored parallel research efforts in **decision-focused learning** and **sequential decision-making under uncertainty** and mentored related research projects, resulting in publications at **AAMAS 2023**, **AAAI 2023**, and EAAMO 2022.
  Awarded a spot bonus for impact in 2021, recognizing "hard work, dedication, talent and creativity".

June–Sept'21 **IBM Thomas J. Watson Research Center**, Yorktown Heights, NY (remote).
RESEARCH INTERN | Manager: Dr. Kush R. Varshney

- Project: Data-Driven Planning and Resource Allocation for Social Change Non-profits **(paper)**

May–July'16 **Sony Corporation**, Tokyo, Japan.
R&D INTERN | Mentor: Yohei Kawamoto

- Project: Sensor fusion and machine learning algorithms for signal quality assessment of PPG signals

May–July'15 **Focus Analytics**, Mumbai, India.
R&D Intern
- Project: Machine learning algorithms for WiFi-based indoor localization & path-tracking

December'14 **Plancess Edusolutions**, Mumbai, India.
Software Engineer
- Project: Algorithms for IIT-JEE rank predictor and study planner/scheduler

Academic Achievements

2023 IAAI 'Innovative Application' Award for deployed AI system in maternal healthcare.

2022 INFORMS "Doing Good with Good OR" Competition Runner-up prize.

2021 **Best Paper Award** at NeurIPS'21 Workshop on Machine Learning in Public Health (MLPH).

2020 **Best Lightning Paper Award** at NeurIPS'20 Workshop on Machine Learning in Public Health.

2020 **Best Paper Award** (theme) at NeurIPS'20 Workshop on Machine Learning for Health (ML4H).

2013 **Selected to represent India** by the Dept.of Science and Technology, Govt.of India, at University of Sydney, Australia and received the Prof. Harry Messel International Science School scholarship.

2013 **All India Rank 273**, amongst 1.3 million examinees in the IIT Joint Entrance Exam (JEE) 2013.

2012 Secured **All India Rank 15** in the KVPY scholarship awarded by Government of India.

2013 Selected among top 300 participants of India to compete in **all three national olympiads**: **INPhO** (Indian National Physics Olympiad), **INChO**(Chemistry), **INAO** (Astronomy).

2009 Selected for the Indian National Mathematics Olympiad **(INMO)** training camp.

2009 Awarded the National Talent Search (NTS) Scholarship, by Government of India.

Publications

Book Chapters

June 2023 **Aditya Mate**, Shresth Verma, Gargi Singh, Aparna Taneja and Milind Tambe
"Field Study in Deploying Restless Multi-Armed Bandits: Assisting Non-Profits in Improving Maternal and Child Health", [link].

Journal Publications

[J5] MSJAR 2025 Tian Xia, **Aditya Mate***, Regina Ceballos*, et al.
"Building Gold-Standard Datasets for AI Quality Evaluation", *Microsoft Journal on Applied Research (MSJAR) volume 22*, *Equal contribution, [MSFT confidential].

[J4] JAAMAS 2025 Feiran Jia, **Aditya Mate**, Zun Li, Shahin Jabbari, Mithun Chakraborty, Milind Tambe, Michael P. Wellman and Yevgeniy Vorobeychik
"A Game-Theoretic Approach for Hierarchical Epidemic Control", *Journal on Autonomous Agents and Multi-Agent Systems, Volume 39, 2025*, [paper].

[J3] MSJAR 2023 Newman Cheng, David Koleczek, Ha Trinh, Kate Lytvynes, **Aditya Mate**, Allen Stewart, Anusha Nandam, Ayleen Durasno, Billie Rinaldi, . . ., Steven Truitt, Chester Curme, Jonathan Larson.
"Graph Based LLM Agents for Complex Reasoning Tasks", *Microsoft Journal of Applied Research (MSJAR) Volume 20, (2023)*, [paper internal to Microsoft].

[J2] PNAS 2020 Bryan Wilder, Marie Charpignon, Jackson A Killian, Han-Ching Ou, **Aditya Mate**, Shahin Jabbari, Andrew Perrault, Angel Desai, Milind Tambe, and Maimuna S. Majumder.
"Modeling between-population variation in COVID-19 dynamics in Hubei, Lombardy, and New York City", *Proceedings of the National Academy of Sciences (PNAS) 2020*, [paper].

[J1] HFES 2020 Palvi Aggarwal, Omkar Thakoor, **Aditya Mate**, Milind Tambe, Edward A. Cranford, Christian Lebiere, and Cleotilde Gonzalez.
"An Exploratory Study of a Masking Strategy of Cyberdeception Using CyberVAN", *Proceedings of the Human Factors and Ergonomics Society Annual Meeting (HFES) 2020*, [paper].

Rigorously Reviewed Conference Publications

[C14] CAIS 2026 Emmanuel Boateng, et al. **Aditya Mate**, Ehi Nosakhare.
"SAPO: Secure Automated Prompt Optimization via Multi-Agent Collaboration", *ACM Conference on AI and Agentic Systems (CAIS) 2026*, [paper].

[C13] EMNLP 2025 Anh Pham, Mihir Thalanki, Michael Sun, Aditya Chaloo, Ankita Gupta, Tian Xia, **Aditya Mate**, Ehimwenma Nosakhare, Soundararajan Srinivasan
"How to Fine-Tune Safely on a Budget: Model Adaptation Using Minimal Resources", *Empirical Methods in Natural Language Processing (EMNLP) 2025 Industry Track*, [paper].

[C12] ICML 2023 **Aditya Mate**, Bryan Wilder, Aparna Taneja and Milind Tambe
"Improved Policy Evaluation for Randomized Trials of Algorithmic Resource Allocation", *International Conference on Machine Learning (ICML) 2023*, [paper].

[C11] AAMAS 2023 Shresth Verma, **Aditya Mate**, Kai Wang, Neha Madhiwalla, Aparna Hegde, Aparna Taneja and Milind Tambe
"Restless Multi-Armed Bandits for Maternal and Child Health: Results from Decision-Focused Learning", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2023*.

[C10] AAAI 2023 Kai Wang*, Shresth Verma*, **Aditya Mate**, Sanket Shah, Aparna Taneja, Neha Madhiwalla, Aparna Hegde, Milind Tambe.
"Decision-Focused Learning in Restless Multi-Armed Bandits with Application to Maternal and Child Care Domain", *AAAI Conference on Artificial Intelligence 2023, Washington DC, USA *Equal contribution*, [paper].

[C9] IAAI 2023 Shresth Verma*, Gargi Singh*, **Aditya Mate**, Paritosh Verma, Sruthi Gorantla, Neha Madhiwalla, Aparna Hegde, Divy Thakkar, Aparna Taneja, Manish Jain and Milind Tambe
"Increasing Impact of Mobile Health Programs: SAHELI for Maternal and Child Care", *Innovative Applications of Artificial Intelligence (IAAI-23), Washington DC, USA*, [paper].

[C8] AAAI 2022 **Aditya Mate***, Lovish Madaan*, Aparna Taneja, Neha Madhiwalla, Shresth Verma, Gargi Singh, Aparna Hegde, Pradeep Varakantham and Milind Tambe.
"Field Study and Deployment of Restless Multi-Armed Bandits for Improving Maternal and Child Health Outcomes", *AAAI Conference on Artificial Intelligence 2022, Vancouver, Canada *Equal contribution*, [paper].

[C7] AAMAS 2022 **Aditya Mate**, Arpita Biswas, Christoph Siebenbrunner, Susobhan Ghosh and Milind Tambe.
"Efficient Algorithms for Finite Horizon and Streaming Restless Multi-Armed Bandit Problems", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2022*, [paper].

[C6] UAI 2022 Zun Li, Feiran Jia, **Aditya Mate**, Shahin Jabbari, Mithun Chakraborty, Milind Tambe and Yevgeniy Vorobeychik.
"Solving Structured Hierarchical Games Using Differential Backward Induction" (Oral Presentation), *Conference on Uncertainty in Artificial Intelligence (UAI) 2022*, [paper].

[C5] AAMAS 2021 **Aditya Mate**, Andrew Perrault and Milind Tambe.
"Risk-Sensitive Interventions in Public Health: Planning with Restless Multi-Armed Bandits", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2021* , [paper].

[C4] NeurIPS 2020 **Aditya Mate***, Jackson Killian*, Haifeng Xu, Andrew Perrault and Milind Tambe.
"Collapsing Bandits and Their Application to Public Health Interventions", *Advances in Neural and Information Processing Systems (NeurIPS) 2020, Vancouver, Canada *Equal contribution*, [paper].

[C3] AAAI 2020 — Andrew Perrault, Bryan Wilder, Eric Ewing, **Aditya Mate**, Bistra Dilkina and Milind Tambe. "End-to-End Game-Focused Learning of Adversary Behavior in Security Games", *AAAI Conference on Artificial Intelligence 2020, New York, USA*, [paper].

[C2] AAMAS 2020 — Kai Wang, Andrew Perrault, **Aditya Mate** and Milind Tambe. "Scalable Game-Focused Learning of Adversary Models: Data-to-Decisions in Network Security Games", *International Conference on Autonomous Agents and Multi-agent Systems (AAMAS) 2020, Auckland, New Zealand*, [paper].

[C1] COMSNETS 2020 — B. Sombabu, **Aditya Mate**, D. Manjunath, Sharayu Moharir. "Whittle Index for AoI-aware scheduling", *In 2020 12th International Conference on Communication Systems and Networks (COMSNETS). IEEE, 2020*, [paper].

Selected Workshop Papers

[W9] NeurIPS 2025 — Roshini Pulishetty*, Mani Kishan Ghantasala*, Keerthy Kaushik Dasoju*, Niti Mangwani*, Vishal Garimella, **Aditya Mate**, Somya Chatterjee, Yue Kang, Ehi Nosakhare, Sadid Hasan, Soundar Srinivasan
"One Head, Many Models: Cross-Attention ROuting for Cost-Aware LLM Selection", *Workshop on Evaluating the Evolving LLM Lifecycle NeurIPS 2025*, **Equal contribution*.

[W8] AAAI 2023 — Paritosh Verma, Shresth Verma, **Aditya Mate**, Aparna Taneja and Milind Tambe
"Decision-Focused Evaluation: Analyzing Performance of Deployed Restless Multi-Arm Bandits", *Workshop on Artificial Intelligence for Social Good (AI4SG) 2023*.

[W7] NeurIPS 2022 — Shresth Verma, **Aditya Mate**, Kai Wang, Aparna Taneja and Milind Tambe. "Case Study: Applying Decision Focused Learning in the Real World", *Workshop on Trustworthy and Socially Responsible Machine Learning, NeurIPS 2022*.

[W6] NeurIPS 2021 — **Aditya Mate***, Lovish Madaan*, Aparna Taneja, Neha Madhiwalla, Shresth Verma, Gargi Singh, Aparna Hegde, Pradeep Varakantham and Milind Tambe.
"Restless Bandits in the Field: Real-World Study for Improving Maternal and Child Health Outcomes", *Workshop on Machine Learning in Public Health (MLPH), NeurIPS 2021, Remote*, **Equal contribution*.
Best Paper Award

[W5] NeurIPS 2020 — Aviva Prins, **Aditya Mate**, Jackson Killian, Rediet Abebe and Milind Tambe. "Incorporating Healthcare Motivated Constraints in Restless Bandit Based Resource Allocation", *Workshop on Machine Learning for Health (ML4H), NeurIPS 2020, Vancouver, Canada*.
Best Thematic Submission

[W4] NeurIPS 2020 — Aviva Prins, **Aditya Mate**, Jackson Killian, Rediet Abebe and Milind Tambe. "Incorporating Healthcare Motivated Constraints in Restless Bandit Based Resource Allocation", *Workshop on Machine Learning in Public Health (MLPH), NeurIPS 2020, Vancouver, Canada*.
Best Lightning Paper

[W3] AAMAS 2020 — **Aditya Mate***, Jackson Killian*, Haifeng Xu, Andrew Perrault and Milind Tambe. "Building Decision Aids for Community Health Workers: Optimizing Interventions via Restless Bandits.", *OptLearnMAS, AAMAS 2020, Auckland, New Zealand*, **Equal contribution*.

[W2] KDD 2020 — **Aditya Mate**, Jackson A. Killian, Bryan Wilder, Marie Charpignon, Ananya Awasthi, Milind Tambe and Maimuna S. Majumder. "Evaluating COVID-19 Lockdown Policies For India: A Preliminary Modeling Assessment for Individual States.", *In ACM SIGKDD 2020 Workshop on Humanitarian Mapping*.

[W1] KDD 2020, & ICML 2020 — Bryan Wilder, Marie Charpignon, Jackson Killian, Han-Ching Ou, **Aditya Mate**, Shahin Jabbari, Andrew Perrault, Angel Desai, Milind Tambe and Maimuna Majumder.
"Bayesian inference of between-population variation in COVID-19 dynamics", *Workshop on Machine Learning for Global Health, International Conference on Machine Learning. 2020.* .

Press Coverage

Nov. 2022 **Harvard SEAS Newsletter**, *"CS student helps NGO launch pilot program"*, [article] [tweet].

August 2022 **Google AI Blog**, *"Using ML to Boost Engagement with a Maternal and Child Health Program in India"*, [article].

May 2020 **Business Insider**, *"Four days of work, followed by 10 days of lockdown could help prevent another wave of infections. Here's how the idea compares to other reopening strategies."*, [article].

May 2020 **Medical Xpress**, *What is the right strategy to limit the spread of COVID-19?*, [article].

May 2020 **Harvard SEAS Newsletter**, *"What is the right strategy to limit the spread of COVID-19? Model simulates the impact of different physical distancing policies on the spread of disease"*, [article].

April 2020 **Sakal Media House**, *"Middle ground for India's lockdown situation"*, [article].

April 2020 **Nature Research Asia**, *"Model finds 'middle ground' for India's lockdown exit"*, [article].

Professional Service

- **Area Chair / Senior PC** for ICLR 2027, AAAI 2027, ICML (2026, 2025), NeurIPS 2026, IJCAI 2026, AAMAS 2025. Reviewer for several top-tier AI research conferences in prior years.

Invited Talks

May 2026 **Invited Talk**, *New York AI Agents Summit*, CUNY, Manhattan, NY.

June 2024 **Invited Talk**, *Microsoft Research New England — ML Ideas Seminar*, Cambridge, MA.

March 2024 **Invited Talk**, *University of Maryland — Multi-Agent Reinforcement Learning Group*.

Feb 2023 **Guest Lecture**, *Carnegie Mellon University — course 10-718: Machine Learning in Practice*.

Teaching experience

Fall 2020 **CS 182: "Artificial Intelligence", Harvard University**, *Head Teaching Fellow*,
Instructors: Prof. Milind Tambe and Prof. Boaz Barak.

- Responsible for teaching class sections, designing exams, assignments, grading and coordinating and distributing responsibilities among other TFs.

Spring 2018 **EE 328: "Digital Communication", IIT Bombay**, *Teaching Assistant*.
Instructor: Prof. Kumar Appaiah

Fall 2017 **EE 225: "Network Theory", IIT Bombay**, *Teaching Assistant*.
Prof. Manoj Gopalkrishnan

Extra-Curricular Activities

Cricket

- Play hard ball cricket tournaments for Harvard Cricket Club, MIT Cricket Team, Boston Gymkhana Sports Club, University of Southern California (2018-2019), IIT Bombay Cricket.
- Team **Captain** for Boston Gymkhana Sports Club and IIT Bombay Hostel Cricket Team in Institute Cricket General Championship
- Won '**Best Fielder**' award in Massachussets State Cricket League (MSCL), 2024
- Won '**Batting award**' in New England Cricket League (NECA), 2024
- Won '**Player of the Tournament**' award in IIT Bombay Cricket General Championship, 2016
- **Highest-bid wicket-keeper batsman** in bidding for Trident Cricket League, IIT-B, 2014

Chess

- Finished among **top five tables** in Bournvita Open Interschool **Chess** Tournament and **got featured in newspaper, Times of India (TOI)** for the same